9.3

5 , # 6(1)

5

$m_1 = 6$, $m_2 = 5$, $m_3 = 10$; $P_1(1,5)$, $P_2(3,-2)$, $P_3(-2,-1)$

$M_x = 30 + (-10) + (-10) = 10$

$M_y = 6 + 15 - 20 = 1$

$m = 21$

$\overline{x} = \frac{M_y}{m} = \frac{1}{21}$

$\overline{y} = \frac{M_x}{m} = \frac{10}{21}$

$\left(\frac{1}{21}, \frac{10}{21} \right)$ //

6

(1) $y = \sqrt{x}$, $y = x$

$x = \sqrt{x}$

$x^2 = x$

$\therefore \quad x = 1, \ 0$

$A = \int_0^1 \sqrt{x} - x \, dx$

$= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{1}{2} x^2 \right]_0^1$

$= \frac{2}{3} - \frac{1}{2}$

$= \frac{4-3}{6} = \frac{1}{6}$

$\overline{x} = \frac{1}{A} \int_0^1 x \left(\sqrt{x} - x \right) dx$

$= 6 \left[\frac{2}{5} \cdot x^{\frac{5}{2}} - \frac{1}{3} x^3 \right]_0^1$

$= 6 \left(\frac{2}{5} - \frac{1}{3} \right)$

$= 6 \cdot \frac{6-5}{15}$

$= \frac{2}{5}$ //

$\overline{y} = \frac{1}{A} \int_0^1 \frac{1}{2} \left(x - x^2 \right) dx$

$= 6 \left[\frac{1}{4} x^2 - \frac{1}{6} x^3 \right]_0^1$

$= 6 \left(\frac{1}{4} - \frac{1}{6} \right)$

$= 6 \cdot \frac{1}{12} = \frac{1}{2}$

$\therefore \left(\frac{2}{5}, \frac{1}{2} \right)$ //